Slicing Long-Form Dialogue into TTS-Ready Utterances

Architecture, Empirical Benchmark, and Signal Conditioning

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Abstract

Long-form dialogue is the most abundant raw material for text-to-speech (TTS) corpora, but it is not directly trainable. This document specifies how continuous dialogue is segmented into short, acoustically clean, non-overlapping utterances for speech corpus preparation. We formulate slicing as a constrained interval-proposal problem, state the structural invariants that every proposal must satisfy, and describe a shared boundary algebra that turns heterogeneous proposals into a valid corpus. Five interchangeable strategies are specified: short-time energy with hysteresis, a recurrent neural VAD, a Gaussian-mixture VAD, Transformer word timestamps, and a semantic-acoustic hybrid, each as an explicit input–processing–output pipeline. We evaluate them on 3.42 h of Japanese game dialogue using reference-free boundary metrics, duration and coverage statistics, and real-time factor. Energy maximizes boundary cleanliness and is the recommended default; the hybrid is the best fit for high-fidelity TTS training.

1 Introduction

The first stage of corpus construction produces long-form audio: game dialogue, animation, audiobooks, or archival recordings, usually with music beds, pauses, and no sentence-level annotation. The second stage, slicing, turns that material into the discrete utterances a TTS model can train on. It is the first step after collection, and it sets a ceiling on everything that follows, because a clip that begins mid-fricative or ends in a click cannot be repaired by a better transcript or a larger model.

A first implementation often applies a fixed-duration window or a conventional voice activity detector (VAD). A fixed window cuts through phonemes and ignores syntax; a frame-level detector answers whether a frame is speech, not whether a region is a trainable utterance. The gap between those two questions is the engineering problem addressed here.

1.1 Failure modes as requirements

Four failure modes dominate corpus preparation and recur as design requirements throughout this document:

1. **Unvoiced-phoneme clipping.** Fricatives and plosives have very low RMS but high-frequency content. A pure energy gate can cut them off, producing utterances that begin or end mid-phoneme. The requirement is an edge that lands in a low-energy region without destroying neighbouring speech.
2. **Transient clicks.** A cut at a sample with non-zero amplitude creates a step discontinuity whose spectrum decays as $O(1/f)$, audible as a click after neural vocoding. The requirement is that the first and last samples of a materialized slice be driven smoothly to zero.

3. **Mid-sentence over-segmentation.** Dramatic pauses of 300–500 ms inside a sentence are shorter than a human listener's notion of a sentence boundary but longer than many VAD hangover windows. The requirement is a semantic notion of a clause, not only an acoustic notion of silence.
4. **Run-on utterances.** Monologues longer than roughly 12 s increase VRAM pressure and the risk of attention collapse or hallucination during training. The requirement is a hard ceiling with balanced subdivision rather than crude truncation.

1.2 Slicing as constrained interval proposal

The system does not solve a per-frame classification problem and then hope that the resulting mask is usable. It proposes a set of half-open intervals that jointly satisfy ordering, non-overlap, a duration contract, edge cleanliness, and—when a semantic model is available—plausibility with respect to punctuation and inter-word gaps. Because these constraints conflict, the architecture exposes several interchangeable strategies instead of declaring a single "best" segmenter.

1.3 Contributions

- A constrained interval-proposal formulation with six explicit invariants that every strategy must preserve (Section 3).
- A shared boundary algebra—padding, midpoint collision, conditional merging, balanced subdivision, and zero-crossing snapping—that gives heterogeneous proposals the same structural guarantees (Section 4).
- Five strategies specified as input–processing–output pipelines behind one interface, including a semantic-acoustic hybrid that composes word-timestamp clauses with RMS-dip micro-refinement (Section 5).
- A reference-free evaluation methodology and a full benchmark on 3.42 h of dialogue (Sections 6 and 7).
- Engineering decisions for lazy loading, device fallback, and deterministic output (Section 8).

1.4 Scope

The document does not derive the physics of sound or the mathematics of the Fourier transform. It focuses on the engineering model implemented by the code: representations, decision rules, invariants, failure handling, and measurement. Training of the segmentation networks is out of scope; all neural components are used as pre-trained inference engines. Human perceptual evaluation and voice-cloning fidelity belong to downstream training and appear here only as extension points. The running example is 3.42 h (12,301 s) of continuous Japanese dialogue.

2 Related Work

2.1 Voice activity detection and acoustic segmentation

Acoustic boundary segmentation has historically relied on time-domain and short-time spectral energy thresholds. Early foundational work by Rabiner and Sambur [1] combined short-time energy with zero-crossing rates to detect utterance endpoints in isolated speech. To accommodate variable noise conditions, statistical model-based VADs were introduced; notably, Sohn et al. [2] proposed a decision-directed likelihood ratio test over Gaussian-mixture models (GMM) in discrete sub-bands, which formed the algorithmic basis of the Google WebRTC VAD [3]. While highly computationally efficient ($< 10^{-3}$ RTF), statistical and energy-based VADs lack phonetic awareness and frequently truncate low-energy unvoiced consonants (fricatives and plosives) or misclassify musical accompaniments as speech. Recent advances have shifted toward deep learning: Silero VAD [4] employs an enterprise-grade recurrent neural network operating on 64-band log-mel filterbanks, achieving high acoustic noise robustness. However,

neural VADs model acoustic continuity rather than linguistic completeness, leading to aggressive pause-bridging and run-on utterances that challenge downstream acoustic models.

2.2 Semantic and ASR-assisted alignment

To align segmentation boundaries with linguistic structure, automated speech recognition (ASR) and forced alignment pipelines have been widely explored. Monolithic Transformer models, such as Whisper [5], predict word-level timestamps directly from cross-attention weights. While these timestamps capture syntactic phrase and sentence structures, attention weights provide soft acoustic localization, frequently causing boundaries to cut through phonemes or transitional coarticulation. Forced alignment frameworks, such as the Montreal Forced Aligner (MFA) [6] and WhisperX [7], combine ASR transcripts with phone-level hidden Markov models (HMM) or phoneme-level CTC models to refine word boundaries. However, forced alignment presupposes an accurate phonemic dictionary, requires language-specific acoustic models, and introduces substantial computational overhead. In contrast, our hybrid approach leverages Whisper's coarse macro-boundaries for semantic coherence while performing zero-crossing-snapped, minimum-energy scans directly on the raw waveform, achieving high acoustic edge cleanliness without phoneme-level dictionary dependencies.

2.3 Speech corpus preparation for text-to-speech

The quality and stability of modern neural TTS systems—including autoregressive tokenizers (e.g., GPT-SoVITS) and flow-matching diffusion architectures (e.g., CosyVoice, VITS)—depend critically on the acoustic cleanliness and duration compliance of the training utterances. Established speech corpora such as LJSpeech [9] and LibriTTS [10] were derived from read-out audiobooks with pre-existing, chapter-aligned text manuscripts, allowing alignment algorithms to operate under constrained search spaces. Conversely, mining continuous speech from unscripted media, game dialogue, and animation represents an in-the-wild challenge characterized by wide dynamic ranges, emotional micro-pauses, background music (BGM), and lack of reference transcripts. Slicing in this setting cannot be treated as isolated frame-level classification; it requires a unified boundary algebra that simultaneously enforces non-overlap, duration constraints, zero-crossing stabilization, and loudness normalization.

3 Problem Formulation and System Design

3.1 Notation

Input waveform $x[n]$, $n = 0, \dots, N-1$, sampled at f_s; frame length M and hop H in samples; frame energy $R[k]$ and its level $20 \log_{10} R[k]$ in dBFS. A segmentation is an ordered set $\mathcal{B} = \{(a_k, b_k)\}_{k=1}^{K}$ of half-open index intervals, with $0 \le a_1 < b_1 \le a_2 < b_2 \le \cdots \le N$. The soft target duration window is $[L_{\min}, L_{\max}] = [2, 12]$ s, the hard floor is $S_{\min} = 0.8$ s, and the maximum kept silence per side is $P_{\max} = 0.4$ s. Boundary tolerance is τ (default 25 ms). RTF denotes processing seconds per audio second. The original sampling rate f_s is preserved for every public boundary; internal model rates are an implementation detail.

A valid proposal set must satisfy

$$0 \le a_1 < b_1 \le a_2 < b_2 \le \cdots \le a_K < b_K \le N \qquad \text{(ordering and non-overlap)} \tag{1}$$

$$S_{\min} \le b_k - a_k \le L_{\max} \qquad \text{(duration contract)} \tag{2}$$

$$\mathrm{EER}(x, a_k) \le \varepsilon, \quad \mathrm{EER}(x, b_k) \le \varepsilon, \quad |x[a_k]|, |x[b_k]| \le \delta \qquad \text{(edge cleanliness)} \tag{3}$$

$$\{a_k, b_k\} \text{ near punctuation or inter-word gaps} \qquad \text{(semantic plausibility, when available)} \tag{4}$$

Coverage, defined as $\sum_k (b_k - a_k)/N$, is desirable but subordinate to (2)–(4). In practice the system maximizes useful coverage subject to these constraints, using a lexicographic preference: semantic correctness first when a semantic model is available, acoustic cleanliness second, duration compliance third. This is why the architecture exposes several strategies instead of a single "best" segmenter:

the constraints conflict, and the correct trade-off depends on the source material and the downstream model.

3.2 The data contract

A TTS corpus is not a bag of audio files; it is a set of records that a trainer can consume without further interpretation. The system therefore treats every produced utterance as a tuple

$$u = (\text{audio},\ \text{text},\ \text{speaker},\ \text{language},\ \text{tags},\ \text{timing},\ \text{quality}),$$

where *timing* contains the original start/end offsets and *quality* contains peak level, RMS level, clipping rate, edge metrics, and a boundary status. The exported dataset list follows the widely used `path|speaker|language|text` convention, so that a downstream trainer can ingest the corpus directly. This contract drives most architectural decisions: every stage must either preserve the tuple or enrich it, and no stage may silently change the meaning of a timestamp.

3.3 Two execution paths

There are two consumers of a segmentation, and they must not be confused (Figure 1). The *production path* slices, fades, normalizes, writes audio, transcribes, tags, and exports metadata. The *benchmark path* slices the same raw audio but evaluates the *raw* cuts before fading and normalization. This separation is deliberate: a fade forces the first and last samples to zero, so auditing a materialized file for a transient step jump mostly verifies the fade, not the quality of the proposed boundary. Measuring raw cuts is the only way to compare proposals fairly.

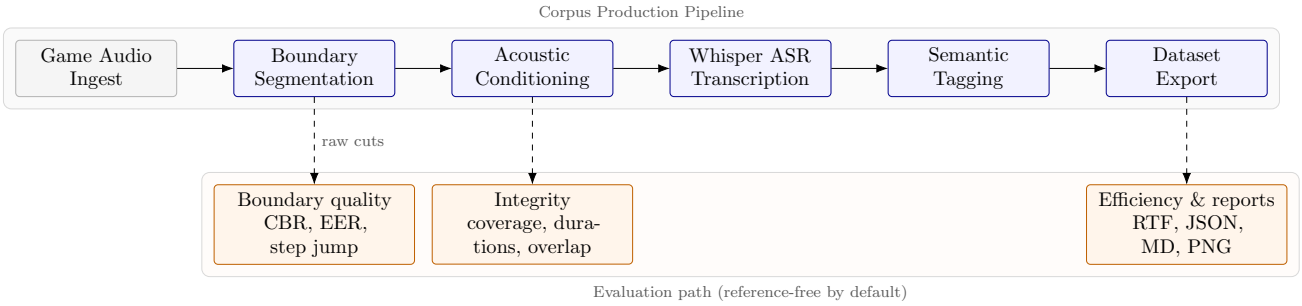


Figure 1: Two execution paths over one segmentation interface. The production path spans interval proposal, acoustic conditioning, transcription, and dataset export; the evaluation path measures raw boundary proposals and system cost. Dashed arrows indicate measurement, not data flow into the corpus.

3.4 Architectural invariants

Six invariants are enforced across all strategies. They are the real interface contract, and every strategy is expected to preserve them:

I1 Coordinate integrity. Every public boundary is an index in the original input sampling rate. Internal resampling for a model is inverted by the exact ratio $f_s/16000$ and clamped to the waveform length.

I2 Non-overlap by construction. For consecutive intervals, $b_k \leq a_{k+1}$. Overlaps introduced by padding are resolved at the midpoint before the interval is emitted; no later deduplication pass is needed.

I3 Duration contract. After subdivision, no interval exceeds $L_{\max}$; intervals below the hard floor $S_{\min}$ are dropped, except for an explicit short-utterance policy in the neural strategy that keeps clips down to 0.8 s.

I4 Conditioning is terminal. Fade and normalization never influence boundary search. They are applied only when a slice is materialized.

I5 Isolation. Each strategy writes to its own output namespace, and a benchmark failure in one strategy does not abort the others.

I6 Reproducibility. DSP paths are deterministic; sorting and tie-breaking are stable. Neural paths are deterministic given the same backend, device, and model version.

4 Method: Signal Conditioning

4.1 Ingest: downmix and rational resampling

Input audio is read as 32-bit float. Multi-channel audio is downmixed by arithmetic mean. If the file rate differs from the configured working rate, the waveform is resampled with a polyphase rational filter: for target rate f_t and source rate f_s, let $g = \gcd(f_s, f_t)$, use upsampling $L = f_t/g$ and downsampling $M = f_s/g$, and apply the anti-aliasing low-pass inherent to the polyphase implementation. This is preferred over naive interpolation because it suppresses aliasing when downsampling from, for example, 48 kHz to 44.1 kHz ($L = 147$, $M = 160$). Model-specific resampling to 16 kHz follows the same rule; boundaries are mapped back by $b_{\text{orig}} = \text{round}(b_{16k} \cdot f_s/16000)$ and clamped to $[0, N]$.

4.2 Framing and short-time RMS

Energy-based analysis uses a zero-copy strided view. The signal is reflect-padded by $M/2$, then framed with

$$R[k] = \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} x^2[kH + m]}, \qquad M = 2H,$$

with $H = 10$ ms by default. The strided view avoids allocating a copy of every frame, so memory stays $O(N)$ while the RMS computation is vectorized. Levels are reported as $20 \log_{10} R[k]$ in dBFS. A reflect padding is used rather than zero padding so that the artificial boundary does not create a spurious energy drop at the first and last frames.

4.3 Zero-crossing snapping

Given a target cut index t, the system searches a window $\mathcal{W} = [t - W, t + W]$ for sign changes of x. Among all indices n with $x[n]x[n + 1] \leq 0$, it selects

$$n^* = \underset{n \in \mathcal{W},\, x[n]x[n+1] \leq 0}{\arg\min} |x[n]|.$$

If no sign change exists, it falls back to the minimum absolute amplitude in the window. The objective is to minimize the transient step jump $|x[n^*]|$ and therefore the high-frequency energy injected by an abrupt cut. The search radius is deliberately scale-dependent: about 50 ms for micro-refinement inside a pause, and about 0.5 s for balanced subdivision of a long utterance.

4.4 Balanced subdivision

An interval of length $L > L_{\max}$ is divided into $m = \lceil L/L_{\max} \rceil$ nearly equal parts with $\ell = \lfloor L/m \rfloor$. The i-th internal target is $t_i = a + i\ell$; each target is snapped to a zero crossing within a 0.5 s window. Compared with a fixed-stride loop, this distributes the remainder across all pieces and avoids a short, low-value tail. The resulting pieces satisfy the ceiling and remain contiguous because each snap point becomes both the end of one piece and the start of the next.

4.5 Edge fading

Materialized slices receive a 5 ms raised-cosine fade:

$$w_{\text{in}}[n] = \tfrac{1}{2}\Big(1 - \cos\frac{\pi n}{F}\Big), \qquad w_{\text{out}}[n] = \tfrac{1}{2}\Big(1 + \cos\frac{\pi n}{F}\Big), \qquad n = 0, \ldots, F-1,$$

where $F = 0.005 f_s$. The first and last samples become exactly zero, which removes the step discontinuity and the associated $O(1/f)$ click. A 5 ms window is long enough to smooth the transient and short enough not to audibly attenuate a plosive onset. Slices shorter than $2F$ are left unchanged because a fade would consume the entire signal.

4.6 Loudness normalization

Two modes are supported. *Peak normalization* scales by $g = \min(p_{\text{target}}/p_{\text{peak}}, g_{\max})$ with $p_{\text{target}} = -1$ dBFS and $g_{\max} = 18$ dB, preventing both clipping and noise-floor explosion. *Loudness normalization* measures integrated loudness according to ITU-R BS.1770-4 / EBU R128, including K-weighting and gating, then applies $\min(L_{\text{target}} - \ell, 18)$ dB of gain and a final peak ceiling at -1 dBFS. Because R128 gating requires roughly 400 ms of material, shorter slices or invalid measurements fall back to peak normalization. The gain ceiling is a deliberate safety choice: a near-silent slice should not be amplified into a wall of noise simply to reach a target loudness.

4.7 Per-slice telemetry

Every slice is measured for duration, peak level, RMS level, and clipping rate. Clipping is counted against a -0.05 dBFS threshold rather than a hard ± 1.0 comparison, so that samples already at the representable ceiling are not missed. Edge telemetry is computed over 25 ms onset and offset windows and is discussed in Section 6. These values are attached to the metadata record and are also available to the enrichment phase.

5 Method: Segmentation Strategies

5.1 Shared boundary algebra

All five strategies emit raw proposals and then apply the same algebra. The algebra is the structural core of the corpus contract:

```
INPUT: raw intervals (a_i, b_i), pad P, merge gap G, ceiling Lmax, floor Smin
1. pad each interval by P on both sides; clip to [0, N]
2. sort by start; for each interval in order:
     if start < previous_end:
         mid = floor((previous_end + start) / 2)
         previous_end = mid; start = mid          # I2: zero overlap
     if (start - previous_end) <= G and (end - previous_start) <= Lmax:
         merge into previous
     else:
         append
3. drop intervals shorter than Smin (unless the short-clip policy keeps them)
4. for every interval longer than Lmax:
         m = ceil(len / Lmax); l = floor(len / m)
         cut at a + i*l, snapped to the nearest zero crossing
OUTPUT: ordered, non-overlapping intervals satisfying the duration contract
```

Two subtleties matter. First, the midpoint collision rule assigns the contested region symmetrically, so neither neighbour is favoured. Second, merging is conditional on the ceiling: two intervals are merged

only if the combined interval still fits $L_{\max}$; otherwise they remain separate and receive the padded boundary. This prevents the common bug where a long pause is bridged into an untrainable monologue.

Each strategy below is documented in the same form: a pipeline table (stage, operation, and parameters) followed by compact properties. The *evidence* stage is what distinguishes the algorithms; ingest, decision, and post-processing differ mainly in parameters, and every path terminates in the shared boundary algebra.

5.2 Energy: short-time RMS with Schmitt-trigger hysteresis

Energy is the DSP baseline. It needs no model, no accelerator, and no training data; its only evidence is loudness relative to a threshold.

Table 1: Energy strategy pipeline.

Stage	Operation	Parameters
Ingest	float32 load, mean downmix, optional polyphase resample	public boundaries stay in the original f_s
Evidence	zero-copy strided RMS on reflect-padded frames, level in dBFS	$M = 2H = 20$ ms, $H = 10$ ms
Gate	dual-threshold Schmitt state machine; adaptive threshold option	$\theta = -38$ dBFS; $T_{\text{hi}} = 10^{(\theta+2)/20}$, $T_{\text{lo}} = 10^{(\theta-4)/20}$; $\theta^* = \text{clip}(P_{10} + 6, -48, -28)$
Cut policy	silent runs: midpoint split, or keep K per side	ignore $d < G = 400$ ms; $K = 400$ ms; drop $S_{\min} < 0.8$ s
Output	shared algebra: pad, merge, subdivide, snap	$L_{\max} = 12$ s; zero-cross radius 0.5 s

- **Strength.** Deterministic and model-free; cuts land in true silence; mean and 95th percentile inside the training window with no run-ons; RTF 5.5×10^{-4}.
- **Limit.** Music above the threshold is treated as speech, and pauses longer than 400 ms become boundaries; 6.8% of the source is discarded as silence or sub-floor fragments.
- **Measured.** CBR 99.6%, mean 3.20 s, coverage 93.2%.

5.3 Silero VAD: recurrent neural voice activity detection

Silero replaces the fixed threshold with a pre-trained recurrent network that scores short log-mel windows and carries acoustic context through a hidden state.

Table 2: Silero strategy pipeline.

Stage	Operation	Parameters
Ingest	float32 load, downmix, resample to 16 kHz	inverse ratio $f_s/16000$ restores coordinates
Evidence	64-band log-mel, approximately 32 ms windows, recurrent layers with hidden state	output $p_t \in [0, 1]$; context suppresses isolated frame errors
Gate	sigmoid threshold plus timestamp state machine	threshold 0.5; min speech 250 ms; min silence 400 ms; padding 400 ms
Output	merge, inverse map, midpoint collision, shared algebra	short-clip policy keeps ≥ 0.8 s; ceiling 12 s

- **Strength.** Robust to music and low-level unvoiced consonants; the 250 ms speech floor preserves short interjections ("Hai!", "Eh?").
- **Limit.** Bridges short pauses: mean duration 8.13 s and 34.9% of clips above 10 s, which is costly for autoregressive training.
- **Cost.** One neural forward pass per window; RTF 0.01178, roughly twenty times the DSP path.

5.4 WebRTC VAD: Gaussian-mixture likelihood ratio

WebRTC is the lightweight statistical baseline: a per-frame Gaussian-mixture likelihood ratio with an explicit aggressiveness setting.

Table 3: WebRTC strategy pipeline.

Stage	Operation	Parameters
Ingest	resample to 16 kHz, 16-bit PCM, fixed frames	10/20/30 ms; 20 ms default
Evidence	six sub-band spectra; speech/noise GMM likelihood ratio per band	aggressiveness mode $\{0, 1, 2, 3\}$; mode 3 rejects the most
Gate	binary frame flags plus 200 ms ring-buffer smoothing	on at $\geq 60\%$ voiced; off at $\leq 20\%$ voiced
Output	intervals, padding, merge, inverse mapping, shared algebra	$P_{\max} = 400$ ms

- **Strength.** Fastest strategy (RTF 0.00082) with a tiny memory footprint and no neural dependency; useful as a baseline and a sanity check.
- **Limit.** The frame-independent GMM confuses fricatives and broadband noise with speech: CBR 63.0% and 37.0% WARN, the worst of the five.
- **Measured.** 60.9% of clips exceed 10 s; high coverage (97.6%) reflects retention by cutting through speech, not quality.

5.5 Whisper: semantic word-timestamp segmentation

Whisper is the only strategy whose primary evidence is linguistic. It decodes word-level timestamps and groups them into clauses, so cuts follow syntax rather than silence.

Table 4: Whisper strategy pipeline.

Stage	Operation	Parameters
Ingest	resample to 16 kHz; optional 5-minute streaming chunks	chunk boundary at RMS dip ± 15 s, zero-cross snapped
Evidence	log-mel spectrogram $\rightarrow$ Transformer ASR with word timestamps	Faster-Whisper (CTranslate2); forced language; internal VAD filter
Grouping	clause rules applied in priority order	sentence punctuation $>$ gap $\geq G$ and length $\geq L_{\min} >$ clause punctuation at $0.75L_{\max} > L_{\max}$
Output	sentence ranges, padding, midpoint collision, shared algebra	chunk offsets added back to source time

- **Strength.** Semantically coherent clauses; no mid-sentence fragmentation at dramatic pauses; mean 4.90 s.
- **Limit.** Timestamps are attention predictions, not measured silence, so cuts can land inside a fricative or coarticulated transition: CBR 78.1%, WARN 21.9%.
- **Cost.** Heaviest pipeline (RTF 0.01813) and the most language-dependent; punctuation conventions drive behaviour.

5.6 Hybrid: semantic macro-boundaries with acoustic micro-refinement

The hybrid keeps Whisper's clause boundaries as macro-anchors and uses DSP only to choose the exact sample inside each predicted pause.

Table 5: Hybrid strategy pipeline.

Stage	Operation	Parameters
Macro	Whisper stage reused unchanged: chunking, decoding, clause grouping	same ASR model and punctuation rules
Evidence	per-gap short-time RMS, deepest minimum inside $[e_k, s_{k+1}]$	20 ms window, 10 ms hop; zero-cross snap
Refine	edge scans and forced splits; midpoint if no gap	$\pm P_{\max}$ at first/last edge; ± 0.5 s around ideal split
Output	refined intervals $\rightarrow$ duration filter and shared algebra	subdivision also snapped to a local RMS dip

- **Strength.** Semantic macro-boundaries plus acoustic micro-cuts: CBR 89.4% (+11.3 points over Whisper) and 99.8% coverage, with only 10.6% WARN.
- **Limit.** Inherits ASR hallucination and timestamp error; refinement can only move inside the predicted gap, so a wrong gap stays wrong.
- **Cost.** ASR plus a small DSP pass; RTF 0.02155, about 19% above Whisper.

5.7 Comparative synthesis

Table 6: Strategy comparison by boundary evidence and engineering trade-off.

Strategy	Evidence	Main strength	Main risk	Cost
Energy	Short-time RMS with 6 dB hysteresis	Deterministic, model-free, cuts in true silence, no run-ons	Threshold sensitive to BGM; 400 ms pauses become cuts	Very low
Silero	Log-mel + recurrent net, probability	Robust to noise and BGM; preserves unvoiced edges and short interjections	Bridges pauses; long utterances; neural runtime	Medium
WebRTC	Six-band GMM likelihood ratio	Real-time CPU baseline; tiny footprint	Fricatives and noise mistaken for speech; high clipping rate	Low
Whisper	Transformer ASR word timestamps + punctuation	Semantic clauses; no mid-sentence fragmentation	Cuts may be acoustically mid-phoneme; heavy and language-specific	High
Hybrid	Whisper macro + RMS-dip micro + zero crossing	Semantic completeness and acoustic cleanliness together	Inherits ASR cost and timestamp errors; bounded by the predicted gap	High

6 Evaluation Methodology

6.1 Three reference-free evaluation axes

A single number cannot rank segmentation strategies, because the objectives are partly orthogonal. The evaluation suite therefore measures three reference-free axes independently:

1. **Acoustic boundary cleanliness**: did the cut land in a quiet region, and is the transient discontinuity small?
2. **Segmentation integrity**: how much speech was preserved, how long are the utterances, and do slices overlap?
3. **Computational efficiency**: how much processing time is required per audio second (RTF)?

6.2 Acoustic boundary cleanliness

For each slice, the first and last 25 ms are compared with the core region. With edge RMS R_e, core RMS R_c, and step jump $\Delta = |x[0]| \vee |x[-1]|$:

$$\text{EER} = \frac{R_e}{R_c + \epsilon}, \qquad \text{EdgeLevel} = 20 \log_{10}(R_e + \epsilon) \text{ dBFS}.$$

A slice is classified by the thresholds in Table 7. The clean boundary rate (CBR) is the percentage of slices that are CLEAN or SAFE; it is the headline metric of the benchmark.

Table 7: Boundary classification thresholds. Both edges must satisfy the level and EER conditions; the step condition is evaluated on the maximum of the two endpoints.

Status	Edge level	EER	Step jump	Interpretation
CLEAN	≤ -35 dBFS	≤ 0.20	≤ 0.02	Cut inside quiet silence
SAFE	≤ -27 dBFS	≤ 0.45	≤ 0.05	Cut inside a margin buffer
WARN	otherwise	otherwise	otherwise	Possible mid-phoneme cut

The aggregate report includes the CLEAN, SAFE, and WARN percentages, CBR, mean and 95th-percentile edge level, mean and 95th-percentile EER, and the maximum step jump. Slices shorter than two edge windows ($2W = 50$ ms) are labelled `TOO_SHORT`; because the hard floor $S_{\min} = 0.8$ s is well above $2W$, this edge-case branch is never triggered in standard pipeline runs.

6.3 Segmentation integrity

For a set of files, let T be the total audio duration and D_k the duration of slice k. The suite reports

$$\text{Coverage} = \min\left(100,\ \frac{\sum_k D_k}{T} \cdot 100\right) \%, \qquad \text{Slices/min} = \frac{K}{T/60},$$

together with mean, median, 95th-percentile, minimum, and maximum duration. Overlaps are counted only between consecutive slices within the same source file, which is the only place where they can occur; the maximum overlap in milliseconds is reported alongside the count. Coverage is capped at 100% so that a numerical artefact cannot produce a nonsensical score.

6.4 Efficiency

The real-time factor is

$$\text{RTF} = \frac{t_{\text{wall}}}{T_{\text{audio}}}.$$

An RTF below 1 means faster than real time. For the neural and ASR strategies the number is hardware- and device-dependent, so it should be read as a relative cost within the same run rather than as an absolute speed claim.

6.5 Benchmark protocol and artifacts

The protocol deliberately mirrors the production path while isolating strategies:

1. Load every source file, downmix, and resample to the working rate.
2. For each strategy, run segmentation on identical inputs and identical optional truncation.
3. Evaluate raw intervals immediately, before fade or normalization.
4. Aggregate integrity across files and boundary statistics across slices.
5. Measure wall-clock time around segmentation only, and compute RTF against total audio duration.

6. Export a JSON summary, a Markdown table, and three PNG figures: executive comparison, acoustic boundary detail, and duration distribution.

A failure in one strategy is caught, reported, and excluded from the comparison; the remaining strategies still complete. Garbage collection and GPU cache clearing run between strategies to prevent cross-run memory contamination.

6.6 Threats to validity

The evaluation is honest about its own limits:

- **CBR is a proxy, not ground truth.** It measures whether an edge is quiet, not whether the phoneme sequence is complete. A cut that removes a final fricative entirely can still score CLEAN.
- **Pre- versus post-conditioning.** Benchmark cuts are raw, whereas exported files are faded and normalized. Fading makes endpoints exactly zero, so a step-jump audit of exported files mostly validates the fade. Absolute edge levels also shift with normalization gain, even though EER is scale-invariant.
- **No human perceptual score.** The suite does not measure naturalness or intelligibility; MOS and voice-cloning fidelity metrics belong to downstream synthesis evaluation.
- **Domain and language.** The empirical results come from one speaker and Japanese dialogue. Punctuation rules, pause statistics, and noise conditions do not transfer unchanged to other languages or recording setups.
- **Coverage is not quality.** A strategy can preserve most of the waveform by cutting through speech. High coverage is only meaningful together with a high CBR and a sane duration distribution.
- **Hardware dependence.** RTF and the relative ranking of neural strategies depend on GPU, driver, and library versions.

Evaluation principle. Boundary cleanliness, duration usability, semantic correctness, and cost are four different questions. A corpus preparation decision should be made from the joint distribution—CBR, duration spread, coverage, and RTF—not from the maximum of any single column.

7 Results: Comparative Benchmark

7.1 Measured behaviour on 3.42 hours of dialogue

Table 8 reproduces the full-corpus benchmark for all five strategies. All runs used the same 12,301 s (3.42 h) of source material and the same working rate; RTF is wall-clock segmentation time per audio second, measured in the same session.

Table 8: Full-corpus benchmark (3.42 h / 12,301 s). Whisper and Hybrid slicing runs evaluated with model_size="base" in float16. CBR is CLEAN+SAFE.

Strategy	Slices	CBR %	Clean %	Warn %	Coverage %	Mean s	P95 s	> 10s %	RTF
Energy	3,582	**99.6**	**98.6**	**0.4**	93.2	**3.20**	**7.21**	**0.7**	**0.00055**
Silero	1,465	96.2	94.4	3.8	96.8	8.13	11.71	34.9	0.01178
WebRTC	1,303	63.0	54.1	37.0	97.6	9.22	11.67	60.9	0.00082
Whisper	2,405	78.1	69.0	21.9	95.7	4.90	9.41	3.5	0.01813
Hybrid	2,424	89.4	86.3	10.6	**99.8**	5.07	9.65	4.1	0.02155

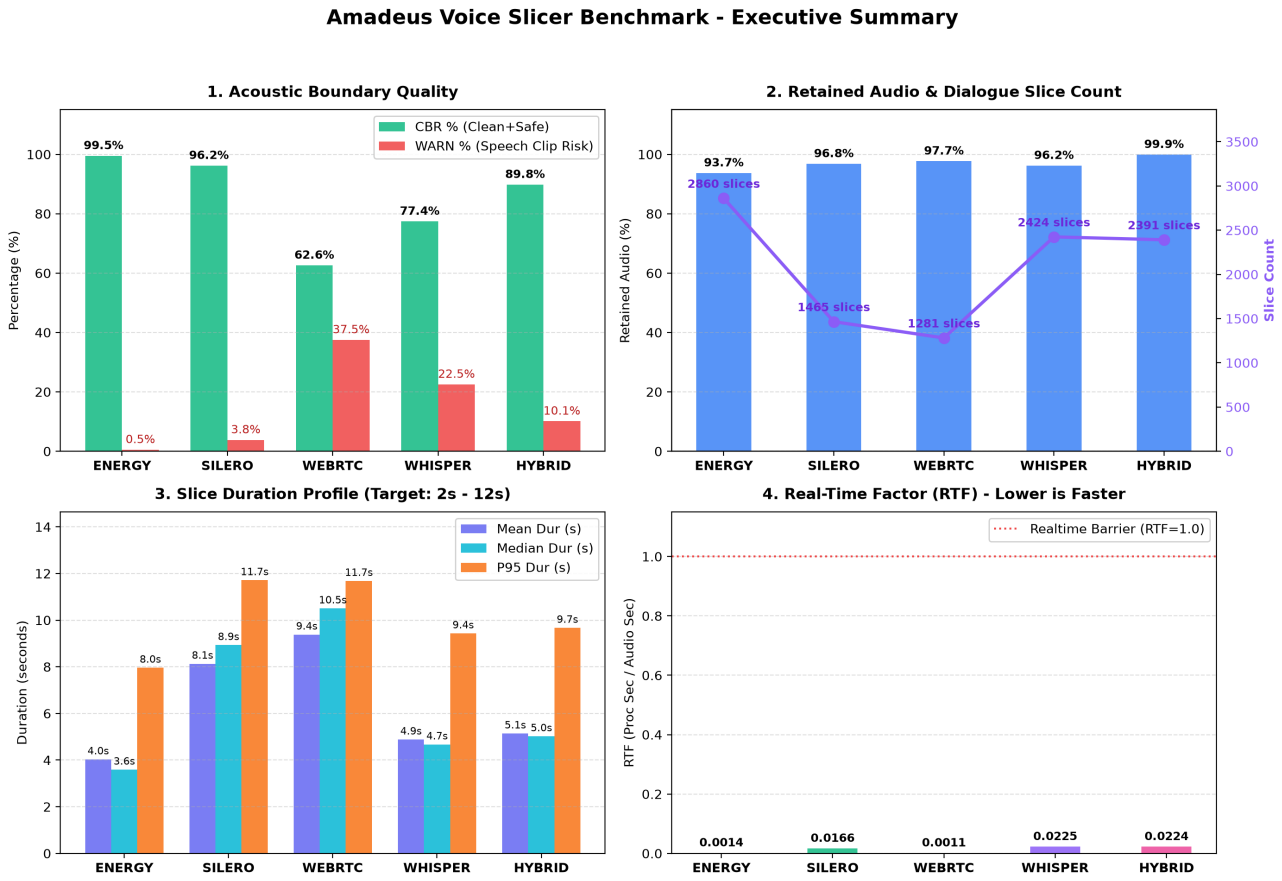


Figure 2: Executive benchmark summary from the same run: boundary quality (CBR versus WARN), coverage and slice count, duration profile, and real-time factor. The complete numerical table is reported above; the chart is generated automatically from the same results.

7.2 Reading the numbers

Energy dominates on boundary quality and duration suitability. Its cuts are placed in silence, so almost every edge is CLEAN (98.6%) and only 0.4% are WARN. Its mean duration (3.20 s, median 2.62 s) and 95th percentile (7.21 s) sit inside the training window, and only 0.7% of clips exceed 10 s. Its RTF of 5.5×10^{-4} is more than three orders of magnitude below real time. The cost is coverage: 93.2% means roughly 6.8% of the source is discarded as silence, noise, or sub-0.8 s fragments. A secondary caveat is the short tail: 30.1% of clips fall below the soft 2 s target, although all remain above the 0.8 s hard floor. For clean dialogue this is an excellent trade, and it is why the strategy is the default recommendation for TTS corpus preparation.

Silero achieves the second-highest boundary score (96.2% CBR) and near-total retention (96.8%), but it bridges short pauses aggressively. Mean duration is 8.13 s (median 8.93 s) and the 95th percentile is 11.71 s; 34.9% of clips exceed 10 s. That is a poor fit for autoregressive TTS training, where long utterances increase memory pressure and attention failure risk. Its boundary score is not evidence of semantic superiority; it is evidence that its cuts, when they happen, land in silence. It remains the right fallback when music or noise defeats a fixed threshold.

WebRTC illustrates the danger of a frame-independent statistical VAD. It preserves 97.6% of the audio but produces 37.0% WARN slices: more than one in three cuts is acoustically suspicious. It also emits the longest clips (median 10.46 s; 60.9% above 10 s). High coverage here is not a virtue; the slicer retains speech by cutting through it.

Whisper produces the most semantically coherent clauses (mean 4.90 s, median 4.69 s, only 3.5% above 10 s), but its word timestamps are not acoustic silence detectors. The 78.1% CBR and 21.9% WARN rate are the measurable consequence: the clause boundary is right, the exact sample often is not. It is the right choice when clause integrity matters more than edge cleanliness, and the wrong choice when the corpus will train a vocoder sensitive to clicks.

Hybrid is the only strategy that improves both axes at once. Reusing Whisper's clauses, it raises CBR from 78.1% to 89.4% and coverage from 95.7% to 99.8%, while keeping the duration profile stable (mean 5.07 s, median 4.94 s, 4.1% above 10 s). The price is an RTF of 0.02155, about 19% above Whisper and roughly 39 times the energy path. The result supports the architectural claim: semantic macro-boundaries and acoustic micro-refinement are complementary, not competing.

7.3 Boundary acoustics in detail

Figure 3 decomposes the acoustic evidence. Energy cuts sit at quiet edges (mean edge level −102.4 dBFS) and combine that with the smallest raw step jump (0.0576, just above the 0.05 click threshold); its mean EER is 0.013. Hybrid is second on both counts (−105.9 dBFS, 0.1765) and improves EER to 0.069 relative to Whisper's 0.162, which is exactly the intended effect of the micro stage. Silero reaches the lowest edge level (−121.1 dBFS) and a low mean EER (0.029), but its maximum step jump is 0.51: a quiet RMS window can still contain a single large sample at the cut. WebRTC has the worst combination (EER 0.266, step 0.69), and Whisper is only marginally better (0.162, 0.69). Because the benchmark measures raw cuts, these step jumps are removed by the 5 ms fade during materialization; they still indicate how much work the fade must do.

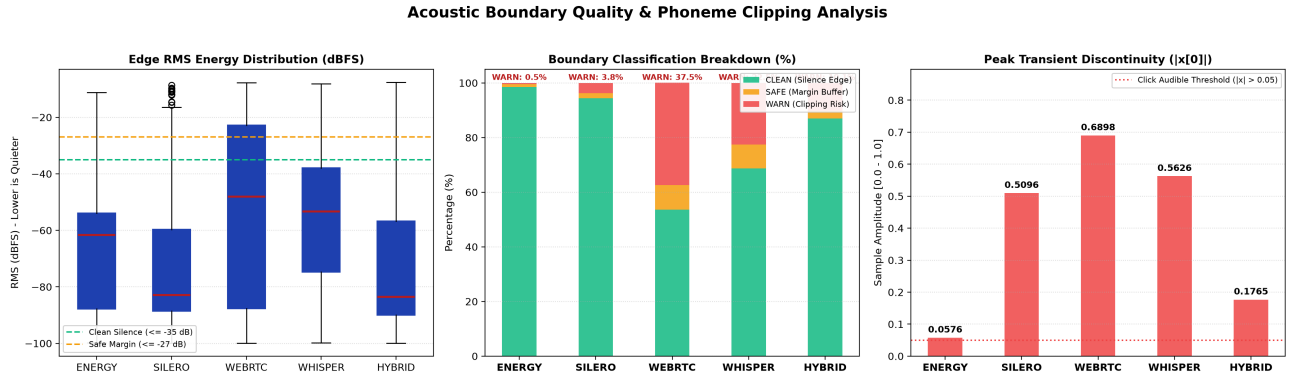


Figure 3: Acoustic boundary detail: edge RMS distribution, CLEAN/SAFE/WARN breakdown, and peak transient discontinuity of raw cuts. The dotted line marks the audible click threshold; conditioning removes this transient in materialized slices.

7.4 Duration distribution

Figure 4 shows that all strategies respect the 12 s ceiling up to zero-cross snapping and padding (empirical maximums: Energy 12.45 s, Silero 11.96 s, WebRTC 11.98 s, Whisper 12.32 s, Hybrid 12.33 s, with over-12 s shares remaining $\leq 0.12\%$), but their mass differs. Energy has the narrowest box (interquartile range roughly 2–4 s) and the lowest median (2.62 s), with a 30.1% short tail; Silero and WebRTC sit high (medians 8.93 s and 10.46 s) with long upper tails; Whisper and Hybrid are compact around 4.7–4.9 s. Measured inside a 2–10 s golden window, Whisper (85.2%) and Hybrid (85.6%) concentrate the most usable utterances, followed by Energy (69.2%), Silero (60.1%), and WebRTC (34.6%). The right panel separates fragmentation (< 2 s) from run-on (> 10 s). Energy has the best run-on rate (0.7%) but the highest fragmentation rate (30.1%). Silero and WebRTC are dominated by run-ons (34.9% and 60.9%). Whisper and Hybrid keep both tails low (11.4%/3.5% and 10.3%/4.1%),

which is the most balanced profile. Fragmentation is cheaper to repair downstream than a run-on, but both consume training slots, so the joint view is the relevant one.

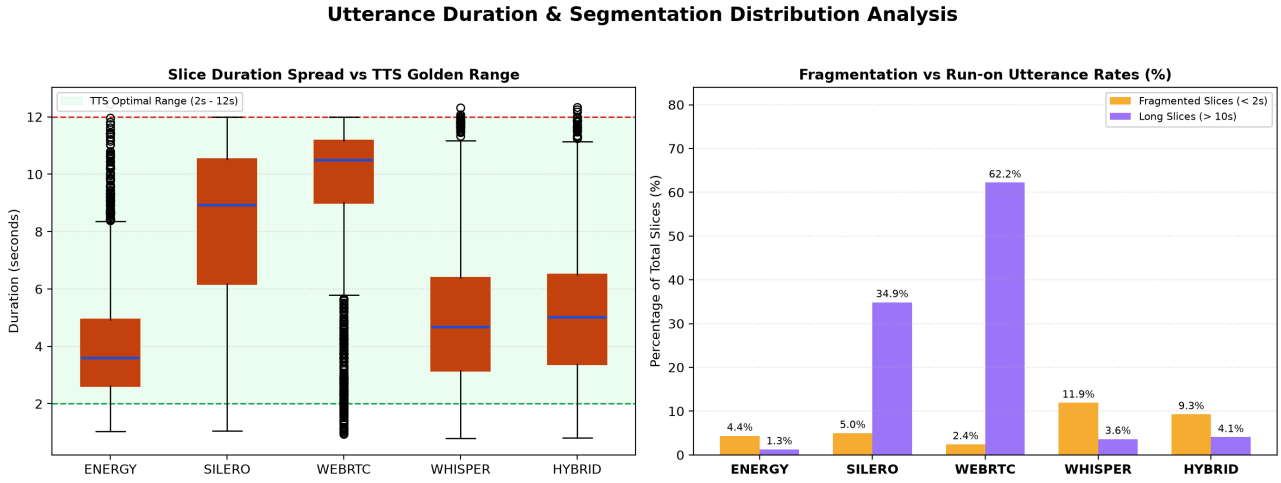


Figure 4: Duration spread against the 2–12 s TTS training band and the fragmentation/run-on rates. A good slicer keeps most mass inside the band and minimizes both tails.

7.5 Recommendation matrix

Table 9: Strategy selection by source material and downstream objective.

Scenario	Recommended	Reason
High-fidelity TTS corpus (GPT-SoVITS, CosyVoice, VITS)	Hybrid (Primary)	Clause integrity by construction, 85.6% in golden range [2s–10s], 99.8% speech coverage, and 89.4% CBR.
High-throughput bulk mining / CPU-only batch	Energy (Secondary)	Highest CBR (99.6%) and zero run-ons at 1,809× realtime speed on CPU; 30.1% sub-2s fragments.
Loud BGM or variable recording level	Silero (or adaptive energy)	Neural VAD separates speech from music; watch for 34.9% run-ons above 10 s.
Semantic baseline without GPU micro-pass	Whisper	Clause-level grouping; accepts higher WARN rate (21.9%) relying entirely on fade window.
Real-time CPU baseline / sanity check	WebRTC	Fast and dependency-light; 37.0% WARN rate disqualifies it for production TTS.

The empirical conclusion is not that one algorithm wins. It is that the same audio can yield boundary lists that are excellent on one axis and poor on another, and that a corpus pipeline must choose its strategy from the joint distribution of CBR, duration, coverage, and cost. The architecture supports that choice by keeping strategies interchangeable and evaluation independent.

8 Runtime Implementation and Reproducibility

8.1 Lazy model loading and device fallback

Neural strategies load their models on first use, not at import time. This keeps the DSP path usable in a minimal environment and avoids paying model-initialization cost for strategies that are not selected. Device selection is defensive: if CUDA is requested but unavailable, the orchestrator falls back to CPU before constructing the model; if a model cannot be moved to the requested device, the strategy falls back to CPU for that model. Compute type follows the device: floating-point half precision on CUDA,

integer quantization on CPU. The principle is that a missing accelerator should reduce speed, not prevent corpus creation.

8.2 Determinism and hardware metadata

DSP operations, sorting, and tie-breaking are deterministic. The benchmark writes a machine-readable summary and a human-readable table from the same in-memory results, so the reported numbers cannot drift between formats. Neural inference is deterministic for a fixed model, backend, device, and driver, but not necessarily across library versions; reports therefore record the strategy, hardware metadata, and configuration rather than assuming bitwise reproducibility across platforms. Output isolation per strategy prevents a later run from overwriting an earlier one and makes A/B comparisons safe.

9 Conclusion

This document has presented the slicing stage of a speech corpus pipeline as a constrained interval-proposal system rather than as a single VAD model. Five strategies propose boundaries from different evidence—energy, recurrent neural activity, Gaussian-mixture likelihood, Transformer word timestamps, and a semantic-acoustic hybrid—but they all obey the same post-conditions: original-rate coordinates, non-overlap by construction, a duration contract, and deterministic conditioning. That shared contract is the architecture. It allows the system to compare radically different algorithms on the same corpus, to swap strategies without changing the rest of the pipeline, and to add new algorithms without touching evaluation.

The evaluation suite separates three core questions that are often conflated: acoustic cleanliness, segmentation integrity, and computational cost. The empirical results show why this separation matters. Energy wins on boundary quality (99.6% CBR) and duration suitability for clean dialogue; Silero is robust but over-bridges pauses, with 34.9% of clips above 10 s; WebRTC preserves audio by cutting through speech, with 37.0% WARN; Whisper produces the most coherent clauses but only 78.1% CBR. The hybrid reconciles the semantic and acoustic criteria, raising Whisper's CBR by 11.3 points and coverage to 99.8% for about 19% more compute. No single strategy dominates every axis, which is precisely why the interface is plural.

The main limitations are explicit and actionable. Reference-free CBR is a proxy for edge quietness, not a proof of phoneme completeness; the slicing stage has no human perceptual score by design; punctuation and pause statistics are language-specific; and coverage can reward a slicer for cutting through speech. The natural next steps for this stage are reference-based evaluation on an annotated subset, true forced alignment for word boundaries instead of attention timestamps, and a streaming variant for corpora larger than memory. Downstream enrichment and model evaluation consume the data contract defined here, attach transcripts and tags, and measure perceptual and speaker-identity quality.

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